Navigating Through Fear and Greed: The Experience-Driven Disposition Effect *

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Abstract

Using transaction-level data on 189,530 Chinese retail investors over the 2013–2016 boom and bust, we measure experience as counts of large realized gains and losses and examine how these counts relate to the disposition effect, the tendency to sell winners and hold losers. A two-standard-deviation increase in losing experience raises the winner–loser sale gap by 23% of the gap at zero measured experience. The corresponding increase in winning experience lowers it by 21%. In calibrated simulations with selling rules that do not respond directly to experience, the mean loss-side interaction is negative or close to zero. We develop a memory model in which similarity-weighted recall guides selling. Realizing gains and losses changes the episodes reached by the ownership cue, linking past outcomes and subsequent selling without changes in preferences.

Keywords: Retail trading, disposition effect, investor experience, memory, cued recall
JEL Codes: D14, D91, G11, G14, G41

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1 Introduction

Trading experience is often expected to reduce investment mistakes. An investor who repeatedly buys and sells securities has opportunities to learn about returns, execution, and her own ability. Studies of the disposition effect—the tendency to sell winners and hold losers—document weaker disposition effects among more experienced investors (Feng and Seasholes, 2005; Seru, Shumway and Stoffman, 2010). Yet a count of trades combines experiences with very different outcomes. Repeated gains may teach a different lesson from repeated losses, even when investors have traded equally often. We ask how these two histories relate to subsequent selling and find opposite associations: accumulated large gains accompany a weaker disposition effect, whereas accumulated large losses accompany a stronger one.

We study transaction records for 189,530 retail investors at a large Chinese brokerage from July 2013 to February 2016. During this period, the market more than doubled and then lost over 40% of its value. The records let us distinguish prior realized outcomes from the unrealized gains and losses on positions still held. We measure winning and losing experience as cumulative counts of sales with returns above 5% and below −5%, respectively, relative to the investor's average purchase price. On dates when an investor sells a security, we observe both the positions sold and those retained, yielding 43.6 million investor–stock–day observations.

In the specification including both experience counts, a two-standard-deviation increase in losing experience is associated with a 0.57-percentage-point increase in the difference between winner and loser sale probabilities. The corresponding increase in winning experience is associated with a 0.51-percentage-point decline. These changes are 23% and 21% of the baseline disposition effect, the fitted 2.48-percentage-point gap at zero measured experience. Investor, stock, and time fixed effects absorb persistent differences across investors and stocks and common market movements. The estimates describe associations as investors accumulate outcomes; the realized gains and losses are not randomly assigned.

The opposite signs persist after trimming extreme experience counts and after controlling for the held stock's recent returns over horizons from one week to six months. They also appear among investors with longer account tenure. The interactions are larger in absolute

value among younger investors. Across holding periods, the pattern appears among positions held for up to 250 days, while neither interaction is precisely estimated for older positions. These comparisons establish where the empirical relationship appears without identifying why investors respond differently to their histories.

An important concern is that experience counts and future selling opportunities change together. A full sale removes a position, and investors with more experience tend to hold more stocks. We therefore simulate portfolio dynamics under random selling, fixed relative selling weights for winners and losers, and a rule that additionally favors positions with extreme returns. The calibration targets the within-investor relationships of experience with portfolio size and the share of winners held. The mean loss-side interaction is negative or close to zero under these rules, unlike the positive empirical estimate. The simulations quantify these particular mechanical explanations; they do not reproduce the full empirical sampling structure or identify a behavioral mechanism.

We also distinguish prior realized outcomes from current portfolio performance. An et al. (2024) show that the disposition effect is weaker when the investor's portfolio is at a gain. We find the same relationship. Adding portfolio gain and its interaction with the position's gain indicator reduces both experience interactions in magnitude. The winning-experience interaction remains negative and significant, while the losing-experience interaction becomes statistically insignificant. Past realized gains therefore contain information about selling beyond the portfolio's current gain status.

We propose memory as one channel linking realized outcomes to later selling. Our model adapts the temporal-context framework of Howard and Kahana (2002) to the choice of a position to sell. The cue "stocks I own" retrieves episodes associated with current ownership, with retrieval depending on similarity and competition among episodes. Realizing a gain or loss changes the mix of outcomes reached by that cue. Past sales can therefore change which outcomes guide a subsequent selling decision without a change in preferences. Connecting this mechanism to the empirical winner–loser sale gap also requires accounting for the numbers of winning and losing positions available to sell.

Our empirical contribution relates first to work on learning and investor sophistication. Feng and Seasholes (2005) and Dhar and Zhu (2006) connect experience and sophistication

to the disposition effect. Nicolosi, Peng and Zhu (2009) document improvements in trading performance with experience. Seru, Shumway and Stoffman (2010) distinguish improvement among continuing investors from the exit of poorly performing investors. Ahn (2020) documents greater learning in the gain domain than in the loss domain, and Kuhnen (2015) shows experimentally that learning from financial information differs across gain and loss contexts. Our comparison concerns the realized outcomes accumulated before a decision: we enter winning and losing experience jointly and examine their opposite associations with the current winner–loser selling gap. It complements evidence on whether investors improve with trading and on how they process favorable and unfavorable information.

The distinction between outcomes also connects to evidence that personal experience shapes financial decisions. Kaustia and Knüpfer (2008) relate personally experienced IPO returns to subsequent IPO subscriptions, and Anagol, Balasubramaniam and Ramadorai (2021) use IPO lotteries to show that gains and losses affect subsequent trading in other stocks even when initial allocations are random. Malmendier and Nagel (2011) link lifetime macroeconomic experiences to risk taking. Imas (2016) shows that realized and paper losses can have different effects on subsequent risk taking. Our outcome is the relative propensity to sell winning and losing positions, rather than participation, overall trading, or the amount of risk taken. This distinction matters because greater willingness to trade need not imply a larger disposition effect.

The paper also relates to explanations of the disposition effect based on preferences and mental accounting. The foundational evidence comes from Shefrin and Statman (1985), Odean (1998), and the trading experiments of Weber and Camerer (1998). Barberis and Xiong (2009) show why prospect theory over annual gains and losses does not reliably generate the disposition effect when initial purchases are considered together with subsequent sales. Their realization-utility framework instead assigns utility to realizing gains and losses (Barberis and Xiong, 2012).[1] Our experience interactions provide an additional relationship for these accounts to explain: how previous realized outcomes relate to the choice between selling a current winner and selling a current loser.

[1]Related analyses examine the role of the decision problem and reference point in prospect-theory explanations (Kaustia, 2010; Hens and Vlcek, 2011; Henderson, 2012; Meng and Weng, 2018).

Mental accounts can extend beyond a single position. Chang, Solomon and Westerfield (2016) link delegation and cognitive dissonance to opposite selling patterns in stocks and mutual funds. Frydman, Hartzmark and Solomon (2018) show that investors can carry a mental account across investments, reducing reluctance to realize a loss when the proceeds are reinvested. Brettschneider, Burro and Henderson (2021) and An et al. (2024) document that the rest of the portfolio matters for position-level selling. These findings motivate our comparison with current portfolio performance and distinguish it from a test of trading skill. They also make clear that a recall mechanism must be considered alongside how investors group and evaluate their investments.

Beliefs provide another explanation for selling behavior. Ben-David and Hirshleifer (2012) document a V-shaped relationship between sale probability and return since purchase, with a steeper gain arm and substantial holding-period variation. They interpret these patterns as consistent with belief revision after large price changes. Andersen et al. (2020) and Qiu et al. (2026) provide further evidence on beliefs and the disposition effect. Jin and Peng (2024) show how a belief in the law of small numbers can reconcile disposition and extrapolation, while Luo et al. (2023) document contrarian trading around news. Our focus is on outcomes realized in earlier sales, rather than only the return on the position currently under consideration. Recent-return controls leave our interactions largely unchanged, but they cannot rule out learning about return dynamics from past outcomes.

The memory interpretation draws on a literature in which the information used in a decision depends on what is recalled. Bordalo, Gennaioli and Shleifer (2020), Malmendier and Wachter (2024), and Wachter and Kahana (2024) connect memory retrieval to economic choice and financial decisions. Gödker, Jiao and Smeets (2025) document experimentally that investors overremember positive outcomes and underremember negative ones. Jiang et al. (2025) link elicited memories to current market conditions, return expectations, and subsequent trading. Enke, Schwerter and Zimmermann (2024) show that associative recall can generate overreaction in beliefs and market outcomes. We apply retrieval to the decision of which holding to sell. Our transaction records contain neither elicited memories nor expectations, so they do not distinguish this route from memory operating through beliefs.

Attention and salient reference points are closely related. Hartzmark (2015) shows that

the best- and worst-performing positions are more likely to be sold, holding their own returns fixed. Frydman and Wang (2020) identify an effect of purchase-price salience on the disposition effect. Quispe-Torreblanca et al. (2025) show that returns since an investor's latest account login matter alongside returns since purchase, while Quispe-Torreblanca et al. (2026) study peak prices experienced during ownership as additional reference points for selling. In recent work, Brettschneider, Burro and Henderson (2026) examine how past best and worst realized returns shape responses to extreme current stock and market returns. We study cumulative counts of large realized gains and losses and their relationship with the current winner–loser sale gap. These are related margins of history-dependent selling, with different measures of experience and different current selling comparisons.

Section 2 describes the data and experience measures. Section 3 presents the empirical results. Section 4 develops the memory model. Section 5 relates the results to known patterns in selling and compares them across investors. Section 6 concludes.

2 Data and Methodology

2.1 Retail Investment Data

We draw on transaction-level records from a major Chinese securities brokerage spanning early July 2013 to late February 2016. The records cover 189,530 individual retail investors trading roughly 4,000 stocks. We use these records to construct the investor–stock–day panel described below. Table 1 reports descriptive statistics. Each record carries the number of shares bought and sold, the purchase and sale prices, and the prevailing market price. Linking records to account IDs adds investor demographics—age, gender, and education.[2] The brokerage also records each investor's mandatory risk-tolerance assessment, completed at account opening.

[2]Under China's real-name registration system, each securities account is tied to a single national identity card, and an investor cannot hold more than one account at the same brokerage. The account we observe therefore reflects the investor's complete holdings at this broker—the relevant unit for measuring the disposition effect and portfolio composition. An investor may in principle maintain a separate account at another brokerage. Unobserved outside positions and trades introduce measurement error both in account-wide portfolio variables and in our experience counts, which capture within-brokerage rather than total personal trading history; the direction of the resulting bias is not determined without further assumptions about the measurement errors and their relation to the regressors.

Figure 1: Chinese Bull and Bear Market, 2013–2016



Notes. The figure plots the daily closing level of the SSE Composite Index against calendar date. The green area marks the July 2013–February 2016 sample period and the gray area the market crash. The index includes all A-shares and B-shares listed on the Shanghai Stock Exchange. Source: Shanghai Stock Exchange.

2.2 Bull and Bear Markets

Figure 1 plots the SSE Composite Index around the sample period. The sample covers bull, bear, and low-volatility periods, allowing us to observe investors' trading histories under different market conditions.[3]

2.3 Constructing the Key Variables

We build our panel at the investor–stock–day level: on each date an investor sells at least one security, we record the positions sold along with every other position the investor holds that day. The regression sample contains 43,649,867 investor–stock–day observations. We

[3]Many relevant studies use predominantly bull-market periods. For example, Feng and Seasholes (2005) in China and Barber and Odean (2000) in the United States both cover sustained upswings. Our 2013–2016 sample spans a full boom and bust, with the index rising from about 2,000 to 5,000 before falling below 3,000.

Table 1: Descriptive Statistics

Panel A: Sample size					
Investors	189,530				
Investor–stock–days	43,649,867				
			Percentiles		
	Mean	SD	25th	50th	75th
Panel B: Investor characteristics					
Account tenure (days)	339.90	162.00	266	308	362
Male investors	55%	-	-	-	-
Investor age (years)	37.88	11.06	29	36	45
Risk-taking level (1–5)	3.05	0.74	3	3	3
Education level (1–7)	4.11	1.28	3	4	5
Panel C: Holdings					
Number of stocks held	7.59	9.24	2.27	4.83	9.52
Portfolio value (RMB)	358,875	3,357,000	19,467	63,472	206,929
Panel D: Trading experience					
Losing experiences	12.40	28.30	0	4	13
Losing experiences (5%)	6.45	13.39	0	2	8
Losing experiences (30%)	0.68	1.92	0	0	1
Winning experiences	14.79	35.95	1	5	16
Winning experiences (5%)	5.66	12.90	0	2	7
Winning experiences (30%)	0.28	1.49	0	0	0

Notes. The sample period is July 2013 to February 2016. The regression sample contains 43,649,867 investor–stock–day observations. Gender and education are available for 128,563 and 122,452 accounts, respectively; analyses using these characteristics use the corresponding subsamples. Risk-taking level ranges from 1 (most risk-averse) to 5 (most risk-seeking). Education level ranges from 1 (PhD) to 7 (junior secondary school or below). Holdings and portfolio value are measured on investor sale dates. The number of stocks held is first averaged across each investor's sale dates; the table reports the distribution of these investor-level averages, which need not be integers. Portfolio values are in renminbi (RMB). Losing and winning experiences count prior sales with negative and positive realized returns; the 5% and 30% measures count returns below the negative or above the positive threshold relative to the investor's average purchase price. Experience and portfolio-value distributions are right-skewed; the maximum Losing Experience count is 2,563.

drop accounts opened before July 1, 2013, our first observation date, so the sample contains only accounts whose complete trading history at this broker we observe. Table 1 reports average account tenure of approximately 340 days.

2.3.1 Portfolio Positions and Investment Gains and Losses

Positions. We define a trading episode as beginning with the first purchase of a stock and ending at its first sale. Shares remaining after a partial sale begin a new episode. If the investor makes additional purchases before selling, the cost basis is the volume-weighted

average purchase price. The realized return is measured relative to this cost basis. Table 1 reports descriptive statistics for holdings and experience. The average investor holds 7.6 stocks, and portfolio values are strongly right-skewed.

Gains and Losses. Following Odean (1998), we evaluate a portfolio's gains and losses only on dates when the investor sells some security, because on no-sale days we cannot tell deliberate holding from inattention (Chang, Solomon and Westerfield, 2016). For each investor–stock–day we compare the relevant price to the cost basis and classify the position as a gain (Gain = 1) when the return is strictly positive and a loss (Gain = 0) when it is strictly negative. Zero-return positions are excluded from the gain–loss comparison. For the stock actually sold, we use the realized return computed at the sell order price; for every other stock held that day, we use the unrealized ('paper') return computed at the day's opening price.[4,5]

A losing experience is a sale with a realized return below a negative threshold, and a winning experience is a sale with a return above the corresponding positive threshold. We count these experiences separately and consider thresholds from 5% to 30%. Without a size threshold, investors record 12 losing and 15 winning experiences on average (Table 1).

Figure 2 plots the account-level distributions of winning and losing experience. Losing experience is more concentrated near zero than winning experience. At higher experience counts, the two distributions are similar. These marginal distributions do not establish whether an investor stopped trading after a loss.

2.3.2 Measuring Trading Experience

Account records avoid survey self-report error but do not reveal how investors perceive their experience. The trade-count measure treats experience as trading activity; the separate winning- and losing-experience measures treat it as realized outcomes.

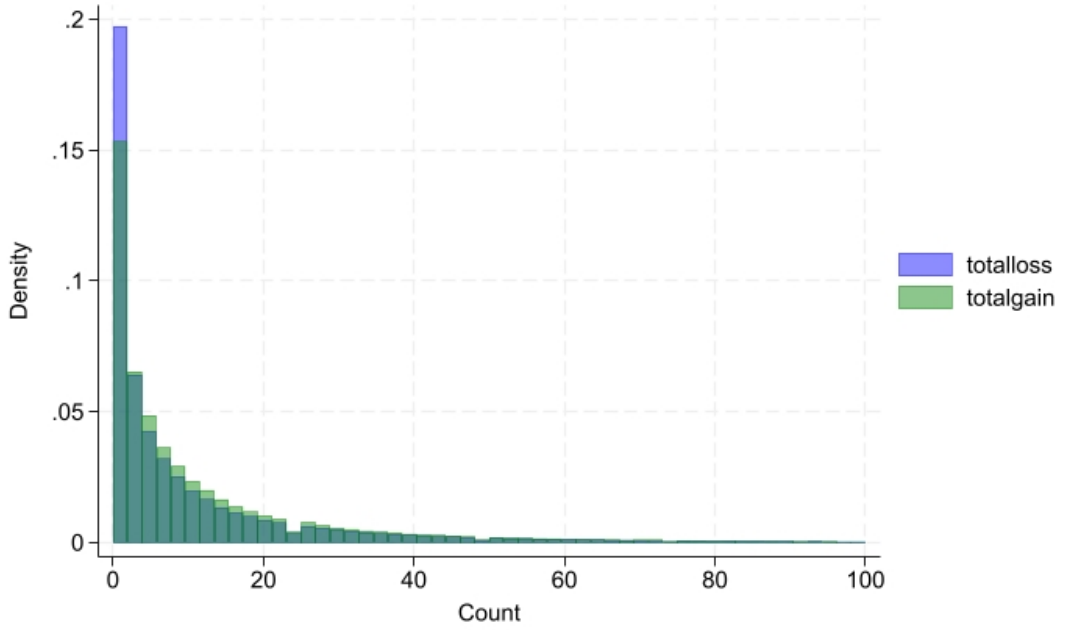
We begin with the trade-count measure of Dhar and Zhu (2006), which proxies experience by accumulated trading activity.[6] Dhar and Zhu (2006) interprets this measure as learning,

[4]We use the opening price because it is predetermined relative to the day's sale decisions; the closing price is not.

[5]Details of our per-transaction return calculations are provided in Appendix A1.

[6]The construction of trading episodes and the associated experience measures is described in Appendix

Figure 2: Distribution of Trading Experience



Notes. The figure shows the distribution of cumulative winning experiences (green, "totalgain") and losing experiences (purple, "totalloss") across accounts at the end of the sample period. The horizontal axis reports the experience count and the vertical axis reports density. In the regressions, experience counts vary over time.

following the experimental evidence in List (2003) that repeated trading increases familiarity and reduces bias. Both studies report a weaker bias among more experienced traders.

We use the pooled trade count as a benchmark. Neither List (2003) nor Dhar and Zhu (2006) distinguishes winning from losing trades, a distinction the sports-card setting of List (2003) does not support. We first estimate the relationship using the pooled count and then distinguish realized gains from realized losses.

Distinguishing Win and Loss Experiences. Let $\mathcal{S}_{i,t}$ denote investor i's sales on dates strictly before t, and let r_e^s be the realized return on sale e. We define

$$\text{Loss Experience}_{i,t} = \sum_{e \in \mathcal{S}_{i,t}} \mathbf{1}\{r_e^s < -X\}$$

A2.

$$\text{Win Experience}_{i,t} = \sum_{e \in \mathcal{S}_{i,t}} \mathbf{1}\{r_e^s > X\}.$$

Here $\mathbf{1}\{\cdot\}$ is an indicator and X is the return threshold. The sums count sales rather than dates, allowing more than one sale on a date. We vary X for robustness and adopt $X = 5\%$ and $X = 30\%$ in our primary analysis.

More active investors have more opportunities to realize both gains and losses. We therefore also divide each experience count by the number of prior sales. For an investor with at least one prior sale, the losing-experience share is

$$\text{Loss Experience Share}_{i,t} = \frac{\text{Loss Experience}_{i,t}}{|\mathcal{S}_{i,t}|}. \tag{1}$$

For example, when $X = 0$, an investor with 15 prior sales, of which 10 realize losses and 5 gains, has $\text{Loss Experience Share}_{i,t} = 2/3$.

2.4 Measuring the Disposition Effect

We measure the disposition effect using an investor-level measure and a position-level regression. For the investor-level measure, we follow Dhar and Zhu (2006) and calculate the difference between the Proportion of Gains Realized (PGR) and the Proportion of Losses Realized (PLR), defined by Odean (1998) as

$$\text{PGR} = \frac{\text{Realized Gains}}{\text{Realized Gains} + \text{Paper Gains}}$$

$$\text{PLR} = \frac{\text{Realized Losses}}{\text{Realized Losses} + \text{Paper Losses}}$$

$$\text{DE} = \text{PGR} - \text{PLR} \tag{2}$$

Realized Gain (Loss) counts the stocks sold in a portfolio at a positive (negative) realized return; Paper Gain (Loss) counts the stocks held at a positive (negative) unrealized return at the time of calculation. Equation 2 is the standard measure of the disposition effect's magnitude: $\text{DE} > 0$ signals the disposition effect, a higher propensity to realize gains than losses.

We also estimate a position-level regression of the sale indicator (1 = Sell; 0 = Hold) on a *Gain* indicator and other controls. Gain is defined using the realized or paper return as described above. Its coefficient measures the fitted difference between winner and loser sale probabilities at the reference values of variables interacted with Gain.

3 Experience-Driven Disposition Effect

Investors in our sample sell winners more often than losers. Across positions observed on sale dates, they realize 12.4% of gains and 8.3% of losses, with an overall realization rate of 9.7%. These unconditional rates are consistent with the disposition effect documented by Odean (1998) and subsequent studies. We next examine how the winner–loser sale gap varies with accumulated experience. The regression separates the fitted gap at zero measured experience from its relationship with subsequent winning and losing experience.

We estimate a linear probability model with investor, stock, and time fixed effects. These absorb persistent differences across investors and stocks and common time variation such as the boom–bust cycle. Standard errors are clustered by investor, day, and stock (Cameron, Gelbach and Miller, 2011), allowing residuals to be correlated within each dimension.

3.1 A Benchmark: Trading Experience and the Disposition Effect

We first estimate the relationship between total trading experience and the disposition effect, following Feng and Seasholes (2005):[7]

$$\begin{aligned} \text{Sale}_{i,j,t} = \alpha + \eta_i + \mu_j + \tau_t + \beta_1 \cdot \text{Gain}_{i,j,t} + \beta_2 \cdot \text{Experience}_{i,t} \\ + \beta_3 \cdot \text{Gain}_{i,j,t} \times \text{Experience}_{i,t} + \epsilon_{i,j,t} \end{aligned} \tag{3}$$

$\text{Sale}_{i,j,t}$ equals 1 if stock j in account i is sold, in whole or in part, at time t, and 0 otherwise. $\text{Gain}_{i,j,t}$ equals 1 if the position has a strictly positive return relative to its average purchase price and 0 if the return is strictly negative, using the price definitions in Section 2.3. Here, α is the common intercept, and η_i, μ_j, and τ_t denote investor, stock, and time fixed

[7] Feng and Seasholes (2005) estimate a hazard model; we estimate the same comparison as a linear probability model.

Table 2: Trading Experience and the Disposition Effect

Dependent variable	Sale × 100	
	Main effects	With interaction
	(1)	(2)
Gain	2.5037***	2.7506***
	(4.74)	(4.77)
Total Experience	0.0025*	0.0029**
	(1.83)	(2.06)
Gain × Total Experience		-0.0020**
		(-2.58)
Constant	8.5853***	8.5345***
	(34.88)	(33.44)
Observations	43,649,867	43,649,867
R-squared	0.139	0.139
Investor, stock, time FE	Yes	Yes

Notes. Linear probability estimates from equation 3. The dependent variable is an indicator for sale, multiplied by 100. Total Experience counts all prior sales, whether at a gain or a loss. Gain indicates that the position is at a gain relative to its cost basis. Column (2) adds the interaction between Gain and Total Experience. The unit of observation is the investor–stock–day; the sample includes holdings on dates when the investor sells at least one security, from July 2013 to February 2016. All specifications include investor, stock, and time fixed effects. Standard errors are clustered by investor, time, and stock; t-statistics appear in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

effects, respectively. At zero measured experience, β_1 is the fitted difference between the sale probabilities of winning and losing positions. We measure experience as

$$\text{Experience}_{i,t} = |\mathcal{S}_{i,t}|.$$

The coefficient of interest is β_3 on the interaction $\text{Gain}_{i,j,t} \times \text{Experience}_{i,t}$. Because *Gain* is binary, experience shifts the sale probability by β_2 for a losing position and by $\beta_2 + \beta_3$ for a winning one, so β_3 measures how the disposition effect varies with experience: a positive coefficient indicates amplification and a negative coefficient attenuation.

Column (2) of Table 2 reports a positive and significant *Gain* coefficient and a negative and significant interaction between *Gain* and *Total Experience*. The disposition effect therefore weakens with accumulated trading experience, consistent with Feng and Seasholes (2005).[8]

[8] Feng and Seasholes (2005) also use experience bins to allow for nonlinearity. Our specification summarizes the relationship with a linear interaction and does not estimate its shape across experience levels.

Table 3: Winning and Losing Experience

Dependent variable	Sale × 100				
	Main effects		With interaction		Joint
	(1)	(2)	(3)	(4)	(5)
Gain	2.4936***	2.4963***	2.4765***	2.6628***	2.4781***
	(4.74)	(4.74)	(4.34)	(4.79)	(4.35)
Loss Experience	0.0163***		0.0162***		0.0109
	(3.17)		(3.12)		(1.13)
Win Experience		0.0087		0.0103	0.0034
		(1.31)		(1.50)	(0.36)
Gain × Loss Experience			0.0007		0.0212***
			(0.30)		(3.28)
Gain × Win Experience				-0.0068***	-0.0198***
				(-2.66)	(-3.30)
Constant	8.4347***	8.6229***	8.4384***	8.5822***	8.4974***
	(37.46)	(35.03)	(36.15)	(33.86)	(36.63)
Observations	43,649,867	43,649,867	43,649,867	43,649,867	43,649,867
R-squared	0.138	0.138	0.138	0.138	0.138
Investor, stock, time FE	Yes	Yes	Yes	Yes	Yes

Notes. Linear probability estimates from equation 3. The dependent variable is an indicator for sale, multiplied by 100. Loss Experience and Win Experience count prior sales with realized returns below −5% and above +5%, respectively, relative to the investor's average purchase price. Gain indicates that the position is at a gain relative to its cost basis. Columns (1) and (3) use losing experience, while Columns (2) and (4) use winning experience; Columns (3)–(4) add the corresponding Gain interaction. Column (5) includes both counts and both interactions. The unit of observation is the investor–stock–day; the sample includes holdings on dates when the investor sells at least one security, from July 2013 to February 2016. All specifications include investor, stock, and time fixed effects. Standard errors are clustered by investor, time, and stock; t-statistics appear in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

3.2 Trading Experience with Both Wins and Losses

Investors with the same number of prior sales can have different histories of gains and losses. We therefore measure winning and losing experience separately. A losing experience is a sale with a return below −5%, and a winning experience a sale with a return above 5%, relative to the investor's average purchase price.[9]

Table 3 reports specifications with separate counts of winning and losing experience.

Column (5) includes winning and losing experience together. A two-standard-deviation increase in losing experience is associated with a 0.57-percentage-point increase in the gap between the probabilities of selling winners and losers. The corresponding increase in win-

[9]We consider thresholds of 5%, 10%, 15%, 20%, 25%, and 30%. We report the baseline estimates at 5% and the estimates using only the largest outcomes at 30%.

ning experience is associated with a 0.51-percentage-point decline.[10] Both interaction coefficients are significant at the 1% level. These differences amount to 23% and 21% of the 2.48-percentage-point *Gain* coefficient, which measures the fitted winner–loser gap at zero measured winning and losing experience.

The winning-experience interaction is negative and significant when included alone in column (4). The losing-experience interaction is small and statistically insignificant when included alone in column (3); it becomes positive and significant when winning experience is also included. The loss result therefore depends on accounting jointly for winning experience. In absolute value, both interaction coefficients in the joint specification are approximately ten times the pooled experience coefficient in Table 2.

To examine sensitivity to extreme experience counts, we re-estimate the baseline specifications after trimming at the 2.5th and 97.5th percentiles. Table 4 reports the results. In the joint specification, the losing-experience interaction remains positive and the winning-experience interaction negative. Both estimates are larger in absolute value than in the baseline.

3.3 Varying the Threshold for Wins and Losses

Figure 3 reports the *Gain* × *Loss Experience* and *Gain* × *Win Experience* coefficients for thresholds from 5% to 30%. The losing-experience interaction is positive and the winning-experience interaction negative throughout. Both increase in absolute value as the threshold rises. Higher thresholds count fewer experiences, however, so the coefficients are measured per rarer event. The larger coefficients do not by themselves establish that more extreme outcomes have greater psychological salience. Table 5 reports the specification at the 30% threshold.

We refer to the larger winner–loser sale gap associated with losing experience as the *fear effect*, and the smaller gap associated with winning experience as the *greed effect*. These labels describe the two patterns. Section 4 explains them through cued recall, and Section 5 examines explanations based on extrapolative optimism and overconfidence.

[10] The two-standard-deviation changes are 26.78 additional losing experiences and 25.80 additional winning experiences.

Table 4: Winning and Losing Experience: Trimmed Sample

Dependent variable	Sale × 100				
	Main effects		With interaction		Joint
	(1)	(2)	(3)	(4)	(5)
Gain	2.5067***	2.5328***	2.4583***	3.1997***	2.8489***
	(4.70)	(4.76)	(3.58)	(4.87)	(4.08)
Loss Experience	0.0368***		0.0361***		-0.0049
	(9.95)		(8.86)		(-1.06)
Win Experience		0.0461***		0.0579***	0.0634***
		(9.45)		(11.80)	(10.03)
Gain × Loss Experience			0.0025		0.0619***
			(0.25)		(4.86)
Gain × Win Experience				-0.0368***	-0.0832***
				(-4.71)	(-10.27)
Constant	8.2791***	8.1548***	8.2945***	7.9312***	8.0765***
	(43.05)	(43.33)	(35.12)	(35.39)	(33.97)
Observations	42,561,204	42,562,559	42,561,204	42,562,559	42,089,393
R-squared	0.138	0.138	0.138	0.138	0.138
Investor, stock, time FE	Yes	Yes	Yes	Yes	Yes

Notes. This table re-estimates the specifications in Table 3 after excluding observations below the 2.5th or above the 97.5th percentile of the relevant experience count. Percentiles are calculated across investor–stock–day observations. Columns (1) and (3) trim Loss Experience; Columns (2) and (4) trim Win Experience; Column (5) trims both, so estimation samples differ across columns. The dependent variable is an indicator for sale, multiplied by 100. Loss Experience and Win Experience count prior sales with realized returns below −5% and above +5%, respectively, relative to the investor's average purchase price. Gain indicates that the position is at a gain relative to its cost basis. Columns (3)–(4) add the corresponding Gain interaction; Column (5) includes both counts and both interactions. The unit of observation is the investor–stock–day; the sample includes holdings on dates when the investor sells at least one security, from July 2013 to February 2016. All specifications include investor, stock, and time fixed effects. Standard errors are clustered by investor, time, and stock; t-statistics appear in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

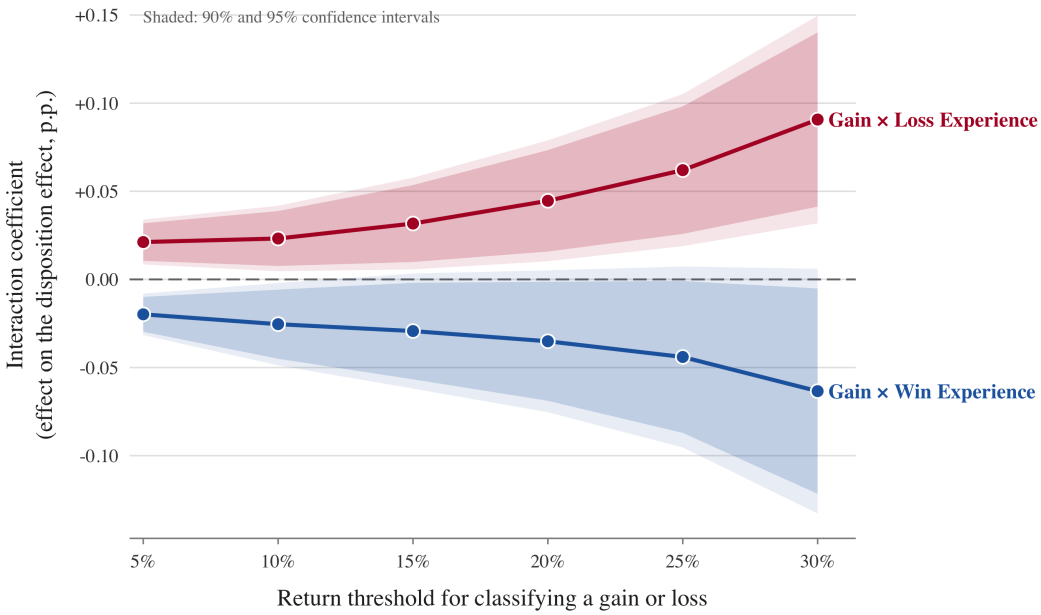
3.4 Trading Experience as a Ratio

More active investors accumulate more winning and losing experiences. We therefore also divide each experience count by the number of prior sales and estimate the model using these shares (Equation 1). This changes the measure from the number of past outcomes to their proportion among prior sales.

Table 6 reports the share specifications at the 30% threshold.[11] The fitted specification

[11] If every prior sale exceeds the chosen threshold in absolute value, the winning and losing shares sum to one. In that case, the two shares are linearly dependent with the intercept, and their interactions with *Gain* are linearly dependent with *Gain*. If the shares nearly sum to one, the joint specification can be ill-conditioned. Estimating both shares jointly requires variation in the share of sales below the threshold.

Figure 3: Experience–Gain Interactions Across Return Thresholds



Notes. The figure plots the *Gain* × *Loss Experience* coefficient (red, fear) and the *Gain* × *Win Experience* coefficient (blue, greed), estimated jointly using the specification in column (5) of Table 3. Main effects are not plotted. The horizontal axis reports the absolute realized-return threshold for counting an experience, from 5% to 30%. Shaded regions show 90% (darker) and 95% (lighter) confidence intervals based on standard errors clustered by investor, time, and stock.

in column (2) implies that increasing the losing-experience share from zero to one widens the winner–loser sale gap by roughly 32 percentage points, compared with a fitted gap of 1.80 percentage points at a zero losing share. This comparison spans the full range of the share, from no prior sales realizing a large loss to all prior sales doing so. The losing-share interaction is positive, while the winning-share interaction in column (4) is negative. When both shares enter together in column (5), both interactions retain their signs and significance.

4 A Memory-Based Model

We present a single mechanism, cued recall combined with skewness in stock returns, that accounts for the baseline disposition effect, its diminution under the experience of gains, and its increase under the experience of large losses. As we explain below, the model can explain further findings regarding selling behavior. That said, we do not claim that it is the only

Table 5: Winning and Losing Experience at the 30% Threshold

Dependent variable	Sale × 100	
	Main effects	With interactions
	(1)	(2)
Gain	2.5063***	2.4275***
	(4.74)	(4.42)
Loss Experience	0.0912*	0.0657
	(1.90)	(1.19)
Win Experience	-0.0035	0.0147
	(-0.04)	(0.14)
Gain × Loss Experience		0.0907***
		(3.02)
Gain × Win Experience		-0.0634*
		(-1.79)
Constant	8.7203***	8.7541***
	(44.35)	(43.88)
Observations	43,649,867	43,649,867
R-squared	0.139	0.139
Investor, stock, time FE	Yes	Yes

Notes. The dependent variable is an indicator for sale, multiplied by 100. Loss Experience and Win Experience count prior sales with realized returns below −30% and above +30%, respectively, relative to the investor's average purchase price. Gain indicates that the position is at a gain relative to its cost basis. Column (1) includes both experience counts; Column (2) adds both Gain interactions. The unit of observation is the investor–stock–day; the sample includes holdings on dates when the investor sells at least one security, from July 2013 to February 2016. All specifications include investor, stock, and time fixed effects. Standard errors are clustered by investor, time, and stock; t-statistics appear in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

explanation for which stocks investors choose to sell, nor that it explains all the findings regarding the disposition effect.[12]

4.1 Selling as Cued Recall

In a cued recall task, a cue brings to mind one of the items associated with it in memory. Retrieved-context theory (Howard and Kahana, 2002) accounts for which item comes to mind through the match between the current context and the context in which each item was encoded, and Wachter and Kahana (2024) apply it to financial decisions. The decision of which stock to sell resembles a cued recall task, with "stocks I own" forming the cue. We use the version of the Howard and Kahana (2002) temporal context model in Kahana, Paron

[12]In particular, Weber and Camerer (1998) and Frydman et al. (2014) report a disposition effect in a laboratory setting with an unconditionally symmetric return distribution.

Table 6: Winning and Losing Experience Shares

Dependent variable	Sale × 100				
	Losing share		Winning share		Joint
	(1)	(2)	(3)	(4)	(5)
Gain	2.4502***	1.8020***	2.4504***	2.5127***	1.8712***
	(4.64)	(3.36)	(4.64)	(4.71)	(3.45)
Loss Experience Share	3.9275***	-3.0334***			-2.9903***
	(5.93)	(-2.89)			(-2.85)
Gain × Loss Experience Share		32.3716***			32.4871***
		(11.63)			(11.66)
Win Experience Share			2.7827***	6.0898***	6.8336***
			(3.45)	(4.67)	(5.29)
Gain × Win Experience Share				-5.9286***	-6.7951***
				(-3.72)	(-4.34)
Constant	8.4407***	8.6346***	8.4973***	8.4649***	8.5648***
	(44.39)	(45.11)	(45.73)	(44.90)	(44.05)
Observations	42,969,625	42,969,625	42,969,625	42,969,625	42,969,625
R-squared	0.127	0.127	0.127	0.127	0.127
Investor, stock, time FE	Yes	Yes	Yes	Yes	Yes

Notes. Linear probability estimates from equation 3 using experience shares defined in equation 1. The dependent variable is an indicator for sale, multiplied by 100. Loss Experience Share and Win Experience Share divide the counts of prior sales with returns below −30% and above +30%, respectively, by the total number of prior sales. Returns are measured relative to the investor's average purchase price. Gain indicates that the position is at a gain relative to its cost basis. Columns (1)–(2) use the losing share and Columns (3)–(4) the winning share, with the corresponding Gain interaction added in Columns (2) and (4). Column (5) includes both shares and both interactions. The unit of observation is the investor–stock–day; the sample includes holdings on dates when the investor sells at least one security, from July 2013 to February 2016. All specifications include investor, stock, and time fixed effects. Standard errors are clustered by investor, time, and stock; t-statistics appear in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

and Wachter (2025).

Agents view features f_i: basis vectors in a high-dimensional vector space. These features become embedded in memory together with a temporal context, an internal mental state that is persistent and evolves through time. Context, through the memory matrix, governs what comes to mind. Specifically, given an initial condition M_0, memory is comprised of outer products of features and contexts $\{f_i, x_i\}_{i=1}^n$:

$$M_n \equiv M_{n-1} + x_n f_n^\top \tag{4}$$

$$= M_0 + \sum_{i=1}^{n} x_i f_i^\top \tag{5}$$

In what follows, we will be interested in a process known as cued recall, which has long been studied in the literature on human memory. Context is retrieved in response to a features cue:

$$\boldsymbol{x}_{\text{CUE}} \propto M_n f_{\text{CUE}}, \tag{6}$$

where context is scaled to have a norm of 1. Substituting (6) into (5) yields:

$$\boldsymbol{x}_{\text{CUE}} \propto M_0 f_{\text{CUE}} + \sum_{i=1}^{n} x_i (f_i^\top f_{\text{CUE}}). \tag{7}$$

Equation 7 shows that features evoke the past contexts under which they are experienced. A context x appears in the sum in (7) if the corresponding f_i equals the current feature; otherwise it does not. Thus context is a weighted average of past contexts under which that feature was experienced. In what follows, we assume M_0 is the zero matrix and we suppress the n subscript on M. Context x_i is endogenous and persistent, and will be described in more detail in what follows.

We use well-established principles of association to determine what pops to mind. As mentioned above, we think of the decision as a cued recall task. In a cued recall task, the cue first retrieves its associated context, as in (6); the context then retrieves features, as in Howard and Kahana (2002):

$$\boldsymbol{f}_{\text{CUE}} \propto M^\top \boldsymbol{x}_{\text{CUE}}. \tag{8}$$

These are features that are experienced under contexts similar to $\boldsymbol{x}_{\text{CUE}}$. There will certainly be nonzero weight on the element corresponding to f_{CUE}, namely $f_{\text{CUE}} \cdot \boldsymbol{f}_{\text{CUE}} \neq 0$. However, any feature experienced at the same time as, or close in time to, f_{CUE} will also be recalled. This is the core mechanism of the temporal context model of Howard and Kahana (2002).

Each candidate feature f_i likewise retrieves a context

$$\boldsymbol{x}_i \propto M f_i. \tag{9}$$

Like $\boldsymbol{x}_{\text{CUE}}$, each retrieved context is scaled to have a norm of one; throughout, the norm is the L^2 norm. This per-candidate normalization is the source of the $1/\sqrt{n_i}$ factors in Theorem 2

below, where n_i denotes the number of observations of f_i.

The vector of retrieved features, $\boldsymbol{f}_{\text{CUE}}$, has nonzero weight on a number of elements. For one of them to rise above the others, it first needs to have a nonzero weight in $\boldsymbol{f}_{\text{CUE}}$, and relatively speaking, a similar context to $\boldsymbol{x}_{\text{CUE}}$. Because recall is competitive, if the context is sufficiently different from $\boldsymbol{x}_{\text{CUE}}$, recall of that feature will be suppressed. The following model captures these properties. The competition runs over feature types rather than individual episodes: multiple episodes may share the same feature, and each of the N distinct feature types stored in memory enters the competition once. For $i = 1, \ldots, N$,

$$\mathbb{P}(\text{recall of } f_i) \propto (\boldsymbol{x}_i \cdot \boldsymbol{x}_{\text{CUE}}) \| f_i \cdot \boldsymbol{f}_{\text{CUE}} \|_0.$$

Equivalently,

$$\mathbb{P}(\text{recall of } f_i) = \frac{(\boldsymbol{x}_i \cdot \boldsymbol{x}_{\text{CUE}}) \| f_i \cdot \boldsymbol{f}_{\text{CUE}} \|_0}{\sum_{j=1}^{N} (\boldsymbol{x}_j \cdot \boldsymbol{x}_{\text{CUE}}) \| f_j \cdot \boldsymbol{f}_{\text{CUE}} \|_0}. \tag{10}$$

Note that $\| f_i \cdot \boldsymbol{f}_{\text{CUE}} \|_0$ is the zero norm; it is either equal to zero or one, zero if f_i is not recalled in (8). This specification is a simplified version of the retrieval process in Lohnas, Polyn and Kahana (2015).

We use the model in Kahana, Paron and Wachter (2025) for context evolution:

$$x_{i+1} \begin{cases} = & x_i \text{ if no distractor} \\ \in & \{x_1, \ldots, x_i\}^{\perp} \text{ otherwise} \end{cases} \tag{11}$$

This formula for context allows for basis vector features that are observed instantaneously to be coded together in context, for example, a stock that is owned is a feature, as is whether the stock has a gain. This allows for the creation of a composite feature, the sum of two feature vectors observed at (nearly) the same point in time. However, (11) is a highly simplified version of what is believed to be true contextual evolution; therefore, we abstract from well-established principles of recall, such as recency and contiguity.

We now link this memory model to the specific context of the decision to sell gains or losses. The agent's information set contains three mutually exclusive categories of items: stocks she owns, stocks she has sold, and other commodities, by which we mean everything

else whose value she observes. Before the agent has any selling experience there are no sold stocks, so $n_{\text{SOLD}} = 0$ and the baseline analysis involves only owned stocks and other commodities. Each item could have either a large or a small return. Define

$$n_c \equiv \text{number of observations of } c$$

and

$$n_{c,c'} \equiv \text{number of joint observations of } c \text{ and } c'$$

for $c, c' \in \{\text{OWNED}, \text{SOLD}, \text{OTHER}, \text{LARGE}, \text{SMALL}\}$. The cue is "stocks I own." Formally, as in Kahana, Paron and Wachter (2025), the cue is a feature: ownership is an attribute of every episode involving a stock the agent holds, in the way that a return category is, and the cue is the feature vector for that attribute, f_{OWNED}. By (6) the cue retrieves the contexts of the episodes that carry it, and with contexts orthogonal across episodes this gives the expressions in Theorem 2 below, with the counts taken over owned-stock episodes. A sale changes the attribute: from the sale on, the episodes of that stock carry SOLD in place of OWNED, so the cue no longer retrieves them, whatever the stock's return category when the episode occurred. We use the category "stocks I own" as shorthand for this cue, and the shorthand notation $\mathbb{P}(\text{LARGE})$ for $\mathbb{P}(\text{ recall of } f_{\text{LARGE}})$, and similarly for $\mathbb{P}(\text{SMALL})$.

The other commodities are the non-stock items whose value the agent observes: cash and deposits, income, the prices of the goods she buys, and the value of a home or a fund if she holds one. Each observation of one of these enters memory as an episode with a return category. At the frequency of a trading decision their changes are small. Individual stocks are the one item the agent observes that routinely moves by the amount that counts as large; the 5% threshold of Section 2 is of that order. Assumption 1 below is the formal version of this claim. The argument requires that the small category contain many non-stock observations. It does not require that any particular non-stock asset class be large, and in particular it does not require holdings of mutual funds or exchange-traded funds.

Assumption 1 *Agents perceive returns as belonging to categories: small and large. Whereas commodities of any type can have small returns, only stocks can have large returns.*

Assumption 2 *The stocks in the agent's memory are the stocks she owns or has sold.*

For a stock, the return the agent records is the gain or loss relative to her purchase price (Section 2), which exists only for a stock she has held. Assumption 1 gives $n_{\textsc{Other,Large}} = 0$ and $n_{\textsc{Other,Small}} = n_{\textsc{Other}}$; Assumption 2 removes stocks the agent has neither owned nor sold from both totals. Together they imply

$$n_{\textsc{Large}} = n_{\textsc{Owned,Large}} + n_{\textsc{Sold,Large}}, \qquad n_{\textsc{Small}} = n_{\textsc{Owned,Small}} + n_{\textsc{Sold,Small}} + n_{\textsc{Other}}. \quad (12)$$

Theorem 1 *Assume that the agent has no winning or losing experiences. The agent competitively recalls either large or small returns with*

$$\begin{aligned} \mathbb{P}(\textsc{Large}) &\propto (\boldsymbol{x}_{\textsc{Large}} \cdot \boldsymbol{x}_{\textsc{Owned}}) \| f_{\textsc{Large}} \cdot \boldsymbol{f}_{\textsc{Owned}} \|_0 \\ \mathbb{P}(\textsc{Small}) &\propto (\boldsymbol{x}_{\textsc{Small}} \cdot \boldsymbol{x}_{\textsc{Owned}}) \| f_{\textsc{Small}} \cdot \boldsymbol{f}_{\textsc{Owned}} \|_0 \end{aligned}$$

with $\mathbb{P}(\textsc{Large}) + \mathbb{P}(\textsc{Small}) = 1$.

Note that we assume that either Large or Small is recalled. Of course, the agent will also recall Owned. As is standard in the memory literature on recall (Lohnas, Polyn and Kahana, 2015), we assume that the agent suppresses the recall of merely repetitive features.

We apply results in Kahana, Paron and Wachter (2025) to connect the right-hand side quantities to the numbers of observations:

Theorem 2

$$\mathbb{P}(\textsc{Large}) \propto \frac{n_{\textsc{Owned,Large}}}{\sqrt{n_{\textsc{Owned}} n_{\textsc{Large}}}} \quad (13)$$

$$\mathbb{P}(\textsc{Small}) \propto \frac{n_{\textsc{Owned,Small}}}{\sqrt{n_{\textsc{Owned}} n_{\textsc{Small}}}} \quad (14)$$

Proof. See Kahana, Paron and Wachter (2025). ■

Corollary 1 *If the number of observations of assets other than stocks is sufficiently large,* $\mathbb{P}(\textsc{Large}) > \mathbb{P}(\textsc{Small})$.

Proof. It follows from Theorem 2 that $\mathbb{P}(\textsc{Large}) > \mathbb{P}(\textsc{Small})$ if and only if

$$\frac{n_{\textsc{Owned,Large}}}{n_{\textsc{Owned,Small}}} > \sqrt{\frac{n_{\textsc{Large}}}{n_{\textsc{Small}}}} \tag{15}$$

By (12) with $n_{\textsc{Sold}} = 0$, $n_{\textsc{Small}} = n_{\textsc{Owned,Small}} + n_{\textsc{Other}}$, whereas $n_{\textsc{Large}} = n_{\textsc{Owned,Large}}$. Increasing $n_{\textsc{Other}}$ increases $n_{\textsc{Small}}$, and hence decreases the right-hand side of (15), while making no other change. The inequality therefore holds for $n_{\textsc{Other}}$ sufficiently large. ■

Assumption 3 *The number of observations of other commodities, $n_{\textsc{Other}}$, is large relative to the number of owned-stock episodes, $n_{\textsc{Owned,Large}} + n_{\textsc{Owned,Small}}$.*

Small price changes are observed far more often than large ones, on stocks and on everything else. Corollary 1 holds as soon as

$$n_{\textsc{Other}} > n_{\textsc{Owned,Small}} \left(n_{\textsc{Owned,Small}} - n_{\textsc{Owned,Large}}\right)/n_{\textsc{Owned,Large}}.$$

Theorem 3 below uses the stronger form, that $n_{\textsc{Other}}$ is large enough for the probability of recalling a small return to be small. Combined with the positive skewness of stock returns, this delivers the baseline disposition effect.

Assumption 4 *The agent sells the stock whose returns pop to mind.*

Assumption 5 *Stock returns are positively skewed.*

In our sample, the skewness of paper returns—unrealized gains and losses at the account level—is 23.1, which confirms a highly right-skewed distribution. Consistent with this, Oh and Wachter (2022) provide broader evidence that stock returns generally exhibit positive skewness.

Assumption 6 *The agent creates memories according to (4) and (11).*

Assumption 7 *Cued recall works as follows. The cue retrieves context $\boldsymbol{x}_{\textsc{Cue}} \propto M f_{\textsc{Cue}}$. This context in turn retrieves the features $\boldsymbol{f}_{\textsc{Cue}} \propto M^{\top} \boldsymbol{x}_{\textsc{Cue}}$. Which of these features gets recalled is competitive with probability determined by nonzero weight in $\boldsymbol{f}_{\textsc{Cue}}$ and similarity of context associated with the features and the cue as in (10)*

Theorem 3 *The agent is more likely to sell gains than losses.*

Proof. By Corollary 1 and Assumption 3, the return that comes to mind is more likely to be large than small. Let π_{Large} and π_{Small} denote the shares of large and of small returns that are gains; by Assumption 5, $\pi_{\text{Large}} > 1/2$. By Assumptions 7 and 4, the agent sells the stock whose return comes to mind, so the stock sold is at a gain with probability $\mathbb{P}(\text{Large})\pi_{\text{Large}} + \mathbb{P}(\text{Small})\pi_{\text{Small}}$. As n_{Other} grows, $\mathbb{P}(\text{Small})$ falls toward zero (the proof of Corollary 1), and this probability approaches π_{Large}, which exceeds one half. The tendency to sell gains is this probability. ■

Large returns are stock-like: they are experienced mostly on stocks, so their contexts match the cue "stocks I own." Small returns are experienced on many things besides stocks, so their contexts are a poorer match, and a small return is less likely to come to mind. Given this structure, the experience effects follow. Let $\mathbb{P}_k(\text{Large})$ denote the probability of recall of large returns, given k winning experiences, and similarly for $\mathbb{P}_k(\text{Small})$.

Theorem 4 *Winning experiences decrease the tendency to sell gains.*

Proof. It follows from Assumption 5 that the large returns are predominantly gains. After k winning experiences $n_{\text{Owned,Large}}$ is replaced by $n_{\text{Owned,Large}} - k$ whereas $n_{\text{Sold,Large}}$ is replaced with $n_{\text{Sold,Large}} + k$. As a consequence, the total number of owned stocks n_{Owned} also falls by k, whereas the total number of sold stocks rises by k. By (12), n_{Large} and n_{Small} are unchanged. It suffices to show that $\mathbb{P}_k(\text{Large})/\mathbb{P}_k(\text{Small})$ is decreasing in k:

$$\frac{\mathbb{P}_k(\text{Large})}{\mathbb{P}_k(\text{Small})} = \frac{n_{\text{Owned,Large}} - k}{n_{\text{Owned,Small}}}\sqrt{\frac{n_{\text{Small}}}{n_{\text{Large}}}}$$

The right-hand side is clearly decreasing in k. ■

Theorem 5 *Losing experiences that are small returns increase the tendency to sell gains.*

Proof. A losing experience of this kind is the sale of a stock with a small return. After k such experiences, $n_{\text{Owned,Small}}$ is replaced by $n_{\text{Owned,Small}} - k$ and $n_{\text{Sold,Small}}$ by $n_{\text{Sold,Small}} + k$; n_{Owned} falls by k, and by (12) n_{Small} and n_{Large} are unchanged. By Theorem 2,

$$\frac{\mathbb{P}_k(\text{Large})}{\mathbb{P}_k(\text{Small})} = \frac{n_{\text{Owned,Large}}}{n_{\text{Owned,Small}} - k}\sqrt{\frac{n_{\text{Small}}}{n_{\text{Large}}}},$$

which is increasing in k. Large returns are predominantly gains, so the tendency to sell gains rises. ■ Selling a small loss removes a small return from the owned category and so strengthens the association between owned stocks and large gains.

Other commodities matter for the baseline disposition effect and not for the direction of the experience results. Corollary 1 requires n_{OTHER} to be large. Theorems 4 and 5 require a weaker condition. Suppose that of the agent's k sales a share λ have large returns and a share $1-\lambda$ small returns, so that $n_{\text{OWNED,LARGE}}$ falls by λk and $n_{\text{OWNED,SMALL}}$ by $(1-\lambda)k$, while by (12) n_{LARGE} and n_{SMALL} are unchanged. By Theorem 2,

$$\frac{\mathbb{P}_k(\text{LARGE})}{\mathbb{P}_k(\text{SMALL})} = \frac{n_{\text{OWNED,LARGE}} - \lambda k}{n_{\text{OWNED,SMALL}} - (1-\lambda)k} \sqrt{\frac{n_{\text{SMALL}}}{n_{\text{LARGE}}}}, \tag{16}$$

which is decreasing in k if and only if

$$\lambda > \frac{n_{\text{OWNED,LARGE}}}{n_{\text{OWNED,LARGE}} + n_{\text{OWNED,SMALL}}}. \tag{17}$$

Selling weakens the recall of large returns whenever large returns are more common among the episodes sold than among the episodes of the stocks owned. Theorems 4 and 5 are the cases $\lambda = 1$ and $\lambda = 0$. Under Assumption 4 the share λ equals $\mathbb{P}(\text{LARGE})$, and (17) then holds if and only if $n_{\text{SMALL}} > n_{\text{LARGE}}$: the experience results require only that small returns outnumber large ones in memory. The size of n_{OTHER} enters the level of the recall probabilities and so the size of the experience effects, and does not affect their direction.

The losing experiences of Section 3 are realized returns below -5%, the same cutoff that makes a gain large, so they are large losses in the model's sense. With two categories of return a large loss would be counted with the large gains, and selling it would work in the same direction as selling a gain. We therefore separate large losses from large gains, introducing feature vectors $f_{\text{LARGE LOSS}}$ and $f_{\text{LARGE GAIN}}$ in place of f_{LARGE} and modifying the notation accordingly.

Like large gains, large losses only occur with stocks. We now need to consider three-way comparisons in the theorems above, but given that large gains are more common than large losses, the baseline disposition effect (Theorem 3), the effect of winning experiences (Theorem 4) and the effect of small losing experiences (Theorem 5) still work under this

extended model.

Theorem 6 *Large losing experiences decrease the probability of recalling a large loss, increase the probability of recalling a large gain, and increase the tendency to sell gains.*

Proof. With three return categories, Theorem 2 gives

$$\begin{aligned}
\mathbb{P}(\textsc{Large gain}) &\propto \frac{n_{\textsc{Owned,Large gain}}}{\sqrt{n_{\textsc{Owned}} n_{\textsc{Large gain}}}},\\
\mathbb{P}(\textsc{Large loss}) &\propto \frac{n_{\textsc{Owned,Large loss}}}{\sqrt{n_{\textsc{Owned}} n_{\textsc{Large loss}}}},\\
\mathbb{P}(\textsc{Small}) &\propto \frac{n_{\textsc{Owned,Small}}}{\sqrt{n_{\textsc{Owned}} n_{\textsc{Small}}}},
\end{aligned}$$

with the three probabilities summing to one. After k large losing experiences, $n_{\textsc{Owned,Large loss}}$ is replaced by $n_{\textsc{Owned,Large loss}} - k$ and $n_{\textsc{Sold,Large loss}}$ by $n_{\textsc{Sold,Large loss}} + k$; $n_{\textsc{Owned}}$ falls by k, and $n_{\textsc{Large loss}}$, $n_{\textsc{Large gain}}$, and $n_{\textsc{Small}}$ are unchanged. The factor $1/\sqrt{n_{\textsc{Owned}}}$ cancels in every ratio, so

$$\begin{aligned}
\frac{\mathbb{P}_k(\textsc{Large gain})}{\mathbb{P}_k(\textsc{Large loss})} &= \frac{n_{\textsc{Owned,Large gain}}}{n_{\textsc{Owned,Large loss}} - k}\sqrt{\frac{n_{\textsc{Large loss}}}{n_{\textsc{Large gain}}}},\\
\frac{\mathbb{P}_k(\textsc{Large gain})}{\mathbb{P}_k(\textsc{Small})} &= \frac{n_{\textsc{Owned,Large gain}}}{n_{\textsc{Owned,Small}}}\sqrt{\frac{n_{\textsc{Small}}}{n_{\textsc{Large gain}}}}.
\end{aligned}$$

The first ratio is increasing in k and the second does not depend on k. Since $\mathbb{P}_k(\textsc{Large gain}) = [1 + \mathbb{P}_k(\textsc{Large loss})/\mathbb{P}_k(\textsc{Large gain}) + \mathbb{P}_k(\textsc{Small})/\mathbb{P}_k(\textsc{Large gain})]^{-1}$, $\mathbb{P}_k(\textsc{Large gain})$ is increasing in k. Since $\mathbb{P}_k(\textsc{Small})/\mathbb{P}_k(\textsc{Large loss})$ equals the first ratio divided by the second, and the second does not depend on k, both ratios in $\mathbb{P}_k(\textsc{Large loss}) = [1 + \mathbb{P}_k(\textsc{Large gain})/\mathbb{P}_k(\textsc{Large loss}) + \mathbb{P}_k(\textsc{Small})/\mathbb{P}_k(\textsc{Large loss})]^{-1}$ are increasing in k, and $\mathbb{P}_k(\textsc{Large loss})$ is decreasing in k. By the same argument $\mathbb{P}_k(\textsc{Small})$ is increasing in k. By Assumptions 7 and 4, the stock sold is at a gain with probability $\mathbb{P}_k(\textsc{Large gain}) + \mathbb{P}_k(\textsc{Small})\pi_{\textsc{Small}}$, where $\pi_{\textsc{Small}}$ is the share of small returns that are gains; both terms are increasing in k. ■

The model's results correspond to the three empirical patterns of Section 3. Theorem 3 delivers the baseline disposition effect with no winning or losing experiences: the returns on the positions an investor holds are positively skewed, so the large returns in memory

are predominantly gains, and the investor recalls, and sells, a gain. Theorem 4 shows that winning experiences decrease the tendency to sell gains; this is the attenuation in *Gain* × *Win Experience*. Theorem 6 shows that large losing experiences increase the tendency to sell gains; this is the amplification in *Gain* × *Loss Experience*, whose losing experiences are realized returns below −5%. Theorem 5 shows that losses that are small work in the same direction.

4.2 Further Implications of Cued Recall

The model speaks to additional findings regarding the disposition effect. First, Ben-David and Hirshleifer (2012) show that the probability of selling a position is V-shaped in its return, which they interpret as an attention effect. While our model is related to attention, the cued recall mechanism is more specific. In our account, large returns of either sign are more stock-like and so come to mind more easily (Corollary 1). The model therefore predicts that these are more likely to be sold. Relatedly, Hartzmark (2015) and Akepanidtaworn et al. (2023) show that individual and institutional investors, respectively, are more likely to sell the best- and worst-performing positions in their portfolios. These are the positions with the most extreme returns, and hence the most stock-like, so they come to mind most easily. Both sets of results are a challenge for explanations based on an asymmetry between gains and losses that does not stem from the properties of returns themselves, such as skewness. Our recall-based model offers a natural explanation.

Second, studies show that the tendency to sell winners disappears or even reverses for mutual funds (Calvet, Campbell and Sodini, 2009; Ivković and Weisbenner, 2009; Chang, Solomon and Westerfield, 2016). One explanation is that the returns are attributed to the fund manager, not to the investor (Chang, Solomon and Westerfield, 2016); another is taxes (Ivković and Weisbenner, 2009). Our model also speaks to this finding: mutual funds are diversified and thus their distribution is far less skewed, so the model predicts a weaker disposition effect for funds.

Finally, laboratory and field evidence point to the importance of the cue itself in shaping decisions, including ownership as a cue (Hartzmark, Hirshman and Imas, 2021). When the purchase price is displayed more prominently, the disposition effect increases (Frydman and Wang, 2020); when it is removed from the screen, the disposition effect falls, and subjects

still act on the gain or loss they remember (Frydman and Rangel, 2014).

5 Additional Results

This section first asks whether the experience effect could arise mechanically, because the portfolio grows with experience and selling does not scale up with the number of positions held. We then include the portfolio's own gain or loss in the regression, ask whether regression to the mean could explain our results, control for the held stock's recent returns at several horizons, and ask whether winning experience raises turnover. We also split the sample into holding-period bins. Finally, we consider the effect of age, which the model relates to the size of the store of past episodes.

5.1 Is the Experience Effect Mechanical?

Investors who have been trading for longer tend to accumulate more positions. If trading activity does not scale with positions, then the probability of selling any one position will fall with the number of positions, and thus also with trading experience. Because disposition is measured by taking the difference between the propensity to sell winners and the propensity to sell losers, the overall size of the effect might diminish with experience due to this effect. Furthermore, if investors who have been trading for longer tend to realize more winners, then the disposition effect would fall, mechanically, with winning experiences. To assess this confound, we turn to simulations.[13]

We conduct a Monte Carlo simulation in which investors sell positions independently of their past experience and estimate an investor-FE analogue of the empirical regression on the simulated data. We calibrate portfolio dynamics to observed moments in the data. First, as the reasoning in the previous paragraph suggests, within-investor experience and portfolio size are strongly positively correlated.[14] It is roughly the same across the two

[13] A more obvious confound may appear to be that as the investor sells winners, there are fewer winners in the portfolio. Under random selling, this reduces the unconditional probability of a winner being sold. It does not, however, reduce the probability of sale conditional on being a winner, which is how the disposition effect is measured.

[14] All our measures are within-investor because our analyses employ investor fixed effects.

types of experiences: +0.61 for winning experience, +0.58 for losing. We also calibrate to the decline in the held winner share as experience accumulates. Winning experience is negatively correlated with the held winner share (−0.26). The held winner share also declines as losing experience accumulates, but less strongly (the correlation is −0.16). Appendix A3 details the data-generating process and plots the simulated distributions. Because selling in the simulation does not depend on experience, any nonzero interaction coefficient in the simulated data is mechanical by construction.

Table 7 reports results under three nulls: random selling, fixed relative selling weights for winners and losers, and the same weights combined with the tendency to sell extreme positions documented by Ben-David and Hirshleifer (2012) ("V-shape selling behavior"). Under random selling, one is not more likely to sell a gain than a loss. Experience correlates with a greater number of positions, but in this simulation there is no baseline disposition effect, and so the propensity to sell a gain over a loss is zero and thus does not fall with experience. This explains the near-zero coefficients on the interaction between gain and both types of experiences.[15] Random selling does not produce the effects we observe: the mean simulated interactions are +0.0006 on the loss side and −0.0022 on the win side.

Under our second null, winners' selling weight is fixed at 1.5 times losers' weight. Indeed, the simulation shows that experience mechanically diminishes the winner–loser selling gap under this rule: losing and winning experiences lead to a lower propensity to sell winners for the reason we describe at the start of this section. This is potentially an issue for interpreting studies focusing on all experiences, rather than the differential effect of winning and losing experiences. Under our calibration, however, the mechanical interactions are small: −0.0064 for losing experience and −0.0032 for winning experience. Because winning experiences and losing experiences correlate with large portfolios about equally, the confound that we are concerned about, namely that winning experience indicates more experience overall, and hence a larger portfolio and a smaller disposition effect, does not explain our results. The reason is that nearly the same dynamics work for losing experiences, resulting in a negative

[15]The win-side coefficient under random selling, −0.0022, is small economically, but too large to be simulation noise. Under equal sale probabilities the true winner–loser gap is zero by construction. As mentioned above, the propensity to sell any one position is decreasing in experience, and does so in a non-linear way. The regression, however, estimates a linear effect. The interaction terms between gain and winning and losing experiences pick up the residual from the misspecification.

coefficient on the interaction, the opposite sign of what we find. Fixed relative selling weights, combined with experience, do not produce the effects we observe either.

Under the second null, the mechanical coefficients are small for both types of experience but less negative for winning experience than for losing experience. This ordering is opposite to the model prediction and the empirical estimates. As emphasized above, the main confound arises from portfolio size. A larger portfolio mechanically leads to a reduced disposition effect. A secondary mechanical effect offsets this main confound: as an investor sells winners, there will be fewer as time goes by, raising the probability of a sale under this selling rule. Winning experiences should thus increase the propensity to sell winners. This portfolio composition effect offsets the portfolio growth effect in our second null, causing the coefficient on the interaction between gain and winning experience to be closer to zero than that between gain and losing experience.

Our third null combines the same relative selling weights for winners and losers with a tendency to sell positions with larger absolute returns that is not modulated by experience. One might think that a tendency to remove the most extreme positions would work roughly equally on the gain and the loss side. Each period, one position is sold with probability proportional to its selling weight. Positions with larger absolute returns receive greater weight, and winners retain a higher weight than losers with the same absolute return. As the number of positions grows, the disposition effect diminishes, though less for winners due to the portfolio composition effect described just above. Table 7 shows, however, that the portfolio composition effect is stronger than in the second null, so much so that the sign on the interaction between gain and winning experience is positive. There are two reasons why the portfolio composition effect is stronger. The first is that, due to skewness, extreme returns tend to be winners. The second is that winners tend to be removed earlier in the simulation for a given investor. The effect of removing winners matters more when there are fewer of them. The net effect is the opposite of what we observe in the data. On the loss side, the coefficient is essentially unchanged from the second null.

5.2 Portfolio Performance and Trading Experience

An et al. (2024) document that the disposition effect is sharply lower when the investor's

Table 7: Monte Carlo Assessment of Mechanical Bias

	Gain × LossExp	Gain × WinExp
Paper estimates (Table 3, Column 5)		
Coefficient	+0.0212	−0.0198
t-statistic	3.28	−3.30
Scenario A: Random selling		
MC mean	+0.0006	−0.0022
95% simulation interval	[−0.020, +0.020]	[−0.024, +0.018]
Tail fraction	0.025	0.040
Scenario B: Fixed relative weights		
MC mean	−0.0064	−0.0032
95% simulation interval	[−0.025, +0.014]	[−0.023, +0.015]
Tail fraction	0.005	0.045
Scenario C: Fixed relative weights + V-shaped selling		
MC mean	−0.0068	+0.0019
95% simulation interval	[−0.025, +0.012]	[−0.013, +0.018]
Tail fraction	0.005	0.005

Notes. This table compares the empirical Gain × Experience coefficients from Table 3, Column (5), with 200 simulations per selling rule at the 5% threshold. Each simulated panel contains 5,000 investors. The simulated regressions use investor fixed effects and investor-clustered standard errors; the empirical regressions also include stock and time fixed effects and three-way clustering. Selling weights do not respond directly to experience, but normalized sale probabilities depend on the current portfolio. MC mean is the average coefficient; the 95% simulation interval is the 2.5th–97.5th percentile range. The tail fraction is the share of simulated coefficients at least as large as the empirical loss interaction or at least as negative as the empirical win interaction. These are descriptive simulation comparisons, not formal p-values for the empirical specification. Appendix A3 describes the calibration and the 30% results. Figure 4 plots the 5% distributions.

portfolio is at a gain. We see a similar pattern in our data (Table 8). One explanation is skill. Skilled investors are more likely to hold portfolios at a gain, and skill may also reduce the disposition effect. Such an explanation suggests a possible confound. More realized gains would indicate more skill, and hence a smaller disposition effect, while more realized losses would indicate less skill, and hence a larger one. Because our experience measures count realized gains and losses, skill could in principle produce both interactions we document. Investor fixed effects do not rule this out, because skill, like experience, can accumulate.

A second possibility does not involve skill. Realized outcomes and the portfolio's current state move together over a boom and bust. An investor who has realized many gains is likely to hold a portfolio at a gain, where the disposition effect is weak, and an investor who has

realized many losses is likely to hold a portfolio at a loss, where it is strong. The interactions in Table 3 could then be the portfolio-level pattern under another name.

We therefore condition on the state of the portfolio. Following An et al. (2024), we add an indicator for whether the portfolio is at a gain, interacted with the position-level gain. With both experience counts included, the regression is

$$\begin{aligned}\text{Sale}_{ijt} = \alpha_i + \mu_j + \tau_t + \beta_G G_{ijt} + \beta_P P_{it} + \delta(G_{ijt}P_{it}) \\ + \beta_L L_{it} + \beta_W W_{it} + \gamma_L(G_{ijt}L_{it}) + \gamma_W(G_{ijt}W_{it}) + \varepsilon_{ijt},\end{aligned} \tag{18}$$

where G_{ijt} indicates that the position is at a gain relative to its cost basis, P_{it} indicates a positive total unrealized return on the account, and L_{it} and W_{it} count prior sales with returns below -5% and above $+5\%$, respectively. Investor, stock, and time fixed effects are denoted by α_i, μ_j, and τ_t. The specification holds the portfolio's state fixed and asks whether past realized outcomes carry information about the winner–loser gap beyond it.

Table 8 reports the estimates. The *Gain* coefficient is the gap when the portfolio is at a loss, 5.3 percentage points, and the gap is 7.9 points smaller when the portfolio is at a gain, so we reproduce the pattern An et al. (2024) document. Holding the state fixed, the losing-experience interaction is 0.0037 and not distinguishable from zero, against 0.0212 in Table 3, Column (5). The winning-experience interaction remains negative and significant, at -0.0097 against -0.0198. The losing-experience association cannot be separated from the portfolio-level effect in this sample. The winning-experience association is not the portfolio-level effect under another name.

5.3 Regression to the Mean

Because experience counts and the investor's past measured disposition effect are both constructed from realized sales, regression to the mean could generate the estimated signs. Suppose every investor has the same true disposition effect and realized selling is noisy around it. An investor who has realized many winners is then, by construction, one whose past measured disposition effect was high by chance, and regression to the mean predicts a lower measured effect going forward; one who has realized many losers had a low past draw and

Table 8: Portfolio Performance and the Experience Effect

Dependent variable	Sale × 100				
	Losing experience		Winning experience		Joint
	(1)	(2)	(3)	(4)	(5)
Gain	5.1801***	5.3538***	5.1827***	5.3958***	5.3288***
	(10.00)	(9.46)	(10.01)	(9.79)	(9.41)
Loss Experience	0.0164***	0.0174***			0.0150
	(3.20)	(3.06)			(1.55)
Gain × Loss Experience		-0.0063			0.0037
		(-1.45)			(0.61)
Win Experience			0.0088	0.0107	0.0015
			(1.32)	(1.53)	(0.16)
Gain × Win Experience				-0.0083**	-0.0097**
				(-2.37)	(-2.37)
Portfolio Gain	2.7078***	2.7270***	2.7019***	2.7064***	2.7019***
	(10.37)	(10.26)	(10.35)	(10.30)	(10.20)
Portfolio Gain × Gain	-7.9363***	-8.0012***	-7.9348***	-7.9617***	-7.9300***
	(-13.26)	(-13.20)	(-13.26)	(-13.22)	(-13.23)
Constant	8.1748***	8.1413***	8.3632***	8.3137***	8.1735***
	(36.54)	(33.93)	(34.29)	(32.84)	(33.98)
Observations	43,649,867	43,649,867	43,649,867	43,649,867	43,649,867
R-squared	0.141	0.141	0.141	0.141	0.141
Investor, stock, time FE	Yes	Yes	Yes	Yes	Yes

Notes. Linear probability estimates from equation 18. The dependent variable is Sale (0/1), multiplied by 100; coefficients are expressed in percentage points. Loss Experience and Win Experience count prior sales with realized returns below −5% and above +5%, respectively, relative to the investor's average purchase price. Gain indicates that the position is at a gain relative to its cost basis. Portfolio Gain indicates a positive total unrealized return on the account at the decision date. All columns include Portfolio Gain and Portfolio Gain × Gain. Columns (1)–(2) use losing experience and Columns (3)–(4) winning experience, with the corresponding Gain interaction added in Columns (2) and (4). Column (5) includes both counts and both interactions. The unit of observation is the investor–stock–day; the sample includes holdings on dates when the investor sells at least one security, from July 2013 to February 2016. All specifications include investor, stock, and time fixed effects. Standard errors are clustered by investor, time, and stock; t-statistics appear in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

should revert upward. Both interactions would follow with no change in behavior at all.[16]

Under this explanation, the relationship between experience and the disposition effect should weaken as longer histories average out noise in past selling decisions. We re-estimate the baseline specification on the subsample of investors with account tenure above the sample median. Table 9 reports the results against the full-sample headline from Table 3, Column (5).

[16]We thank Stefano Cassella for raising this possibility.

Table 9: Trading Experience in the Longer-Tenure Sample

Dependent variable	Sale × 100	
	Full sample	Longer-tenure sample
	(1)	(2)
Gain	2.4781***	1.5401***
	(4.35)	(2.64)
Loss Experience	0.0109	−0.0101**
	(1.13)	(-2.33)
Win Experience	0.0034	0.0291***
	(0.36)	(5.00)
Gain × Loss Experience	+0.0212***	+0.0302***
	(3.28)	(4.37)
Gain × Win Experience	−0.0198***	−0.0219***
	(-3.30)	(-5.31)
Constant	8.4974***	7.7038***
	(36.63)	(35.75)
Observations	43,649,867	28,734,001
R-squared	0.138	0.137
Investor, stock, time FE	Yes	Yes

Notes. Column (1) reproduces Table 3, Column (5). Column (2) uses investors with account tenure above the sample median. The dependent variable is an indicator for sale, multiplied by 100. Loss Experience and Win Experience count prior sales with realized returns below −5% and above +5%, respectively, relative to the investor's average purchase price. Gain indicates that the position is at a gain relative to its cost basis. The unit of observation is the investor–stock–day; the sample includes holdings on dates when the investor sells at least one security, from July 2013 to February 2016. All specifications include investor, stock, and time fixed effects. Standard errors are clustered by investor, time, and stock; t-statistics appear in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

The data move the opposite way. Neither interaction weakens in the long-history subsample, and both are larger in absolute value: *Gain* × *Loss Experience* rises from +0.0212 ($t = 3.28$) to +0.0302 ($t = 4.37$), and *Gain* × *Win Experience* moves from −0.0198 ($t = -3.30$) to −0.0219 ($t = -5.31$), both significant at the 1% level. Regression to the mean predicts attenuation where the noise in past measured behavior has had the most chance to average out; the estimates do not attenuate. Account tenure in this sample varies over months, not decades (interquartile range 266–362 days, Table 1), so this split gauges the stability of the effect within short histories; it does not measure the size of an investor's lifetime store of competing memories, the margin the age test in Section 5.7 addresses.

5.4 Extrapolation and Recent Returns

Recent stock returns may be correlated with realized gains and with investors' selling decisions. If investors extrapolate these returns, the negative interaction between winning experience and *Gain* could reflect return expectations rather than recall of past realized outcomes.

We re-estimate the full 5% specification adding the held stock's cumulative return over the prior week, month, three months, or six months, one horizon at a time. Table 10 shows little change in the experience interactions. Across the four horizons, one additional losing experience is associated with a 0.0200–0.0218-percentage-point increase in the winner–loser selling gap, and one additional winning experience with a 0.0192–0.0200-percentage-point decrease.

Recent stock returns are positively associated with selling at all four horizons, but this association does not distinguish return extrapolation from the V-shaped selling pattern documented by Ben-David and Hirshleifer (2012). Nor do the return controls distinguish recall from all belief-based explanations. Enke, Schwerter and Zimmermann (2024) show experimentally that cued recall of past news can generate overreaction in beliefs and prices. Recall could therefore affect selling through expectations as well as through the retrieval of past experiences. Distinguishing these channels would require belief measures that our data do not contain.

Table 10: Trading Experience, Controlling for Recent Returns

Dependent variable	Sale × 100				
Return-control horizon	Baseline	1 week	1 month	3 months	6 months
	(1)	(2)	(3)	(4)	(5)
Gain	2.4781***	2.2190***	2.1403***	2.2428***	2.1428***
	(4.35)	(3.80)	(3.70)	(3.97)	(3.79)
Loss Experience	0.0109	0.0112	0.0116	0.0118	0.0116
	(1.13)	(1.17)	(1.21)	(1.23)	(1.22)
Win Experience	0.0034	0.0033	0.0030	0.0029	0.0025
	(0.36)	(0.34)	(0.32)	(0.30)	(0.26)
Gain × Loss Experience	0.0212***	0.0218***	0.0216***	0.0210***	0.0200***
	(3.28)	(3.37)	(3.36)	(3.29)	(3.19)
Gain × Win Experience	-0.0198***	-0.0199***	-0.0200***	-0.0197***	-0.0192***
	(-3.30)	(-3.35)	(-3.36)	(-3.33)	(-3.30)
Past-week return		5.4957*			
		(1.75)			
Past-month return			4.0379***		
			(3.04)		
Past-three-month return				2.2432***	
				(3.37)	
Past-six-month return					2.4457***
					(12.47)
Constant	8.4974***	8.3418***	8.3255***	8.0993***	7.1599***
	(36.63)	(36.05)	(36.23)	(33.90)	(29.66)
Observations	43,649,867	42,526,853	42,526,853	42,526,853	42,526,853
R-squared	0.138	0.138	0.138	0.138	0.138
Investor, stock, time FE	Yes	Yes	Yes	Yes	Yes

Notes. Column (1) reproduces Table 3, Column (5). Columns (2)–(5) each add the held stock's cumulative return over the horizon shown in the header. The return-control sample is smaller than the baseline sample. The dependent variable is an indicator for sale, multiplied by 100. Loss Experience and Win Experience count prior sales with realized returns below −5% and above +5%, respectively, relative to the investor's average purchase price. Gain indicates that the position is at a gain relative to its cost basis. The unit of observation is the investor–stock–day; the sample includes holdings on dates when the investor sells at least one security, from July 2013 to February 2016. All specifications include investor, stock, and time fixed effects. Standard errors are clustered by investor, time, and stock; t-statistics appear in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table 11: Winning Experience and Monthly Turnover

Dependent variable	Monthly Turnover	
	(1)	(2)
Win Experience	-0.0825***	-0.2911***
	(-10.14)	(-17.83)
Constant	7.3127***	8.4999***
	(116.51)	(82.97)
Observations	881,576	838,399
R-squared	0.001	0.313
Investor FE	No	Yes

Notes. The dependent variable is investor-level monthly turnover: total purchase plus sale value during the month divided by twice the investor's average portfolio value, (buy + sell)/(2 × average portfolio value). Win Experience counts prior sales with realized gains exceeding 5%. Column (1) excludes and Column (2) includes investor fixed effects. The unit of observation is the investor–month; the sample period is July 2013 to February 2016. Standard errors are clustered by investor; t-statistics appear in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

5.5 Trading Intensity and Overconfidence

An alternative to our memory-based interpretation is that the weaker disposition effect associated with accumulated winning experience reflects gains-induced overconfidence. Investors may attribute successful outcomes to their own trading ability, become more confident, and increase their trading activity (Barber and Odean, 2000; Statman, Thorley and Vorkink, 2006). We examine this implication by testing whether winning experience is positively associated with turnover.

We regress each investor's monthly turnover—the combined RMB value of purchases and sales scaled by twice average portfolio value—on cumulative winning experience, with investor fixed effects and standard errors clustered by investor. Table 11 shows a negative association. Turnover measures the value traded relative to portfolio size, so this result need not imply fewer sales. It weighs against an overconfidence account that predicts higher turnover, while leaving open changes in confidence that affect other trading decisions.

5.6 Holding Period and the Experience Effect

Ben-David and Hirshleifer (2012) emphasize the importance of holding period return in the decision to sell. They find that the longer stocks are held, the less likely they are to be sold,

and that the V-shaped result mentioned earlier falls as holding period lengthens. Holding period and experience are not easy to disentangle fully. However, our study design allows us to distinguish between positions held for different lengths of time.

We examine whether the experience asymmetry persists among positions with different holding periods. We split investor–stock–day observations into three groups by holding period at the decision date: 1–60 days, 61–250 days, and more than 250 days.[17] We re-estimate the baseline regression within each group, allowing the gain coefficient and experience interactions to differ across groups while retaining investor, stock, and time fixed effects and three-way clustering.

Table 12 shows positive loss-experience interactions and negative winning-experience interactions in both the 1–60-day and 61–250-day groups. All four are significant at the 5% or 10% level. The pattern therefore extends beyond the shortest-held positions, although both interaction estimates are smaller in absolute value in the 61–250-day group.

For positions held for more than 250 days, neither interaction is statistically distinguishable from zero. One possible explanation is that investors give little attention to these holdings even while trading other securities, weakening the association between experience and the decision to sell them. Our sample observes such positions on dates when the investor sells at least one security, so this explanation concerns individual holdings within active accounts. Holding duration does not measure attention directly, and earlier decisions not to sell select positions into the oldest group. The regressions cannot distinguish inattention from this selection.

5.7 Trading Experience and Investor Age

Age is a natural proxy for the number of past episodes an investor holds in memory. Through this proxy, our model makes a prediction regarding age: the experience interactions should be stronger among younger investors, who have fewer episodes in memory and therefore a larger share of them represented by each realized outcome (Theorem 2).

[17] Our panel records an investor's holdings on each date on which the investor sells at least one security. An account–stock pair can contribute repeated sale observations through partial sales and non-sale observations when other stocks are sold. Holding period is measured at each decision date and can therefore differ across observations of the same account–stock pair.

Table 12: Trading Experience by Holding Period

Dependent variable	Sale × 100					
Holding period	1–60 days		61–250 days		>250 days	
	(1)	(2)	(3)	(4)	(5)	(6)
Gain	0.6182	0.5655	1.5801***	1.5738***	0.2400	0.2226
	(0.73)	(0.66)	(8.49)	(8.27)	(1.41)	(1.29)
Loss Experience	0.0275	-0.0034	-0.0169***	-0.0211***	-0.0010	-0.0005
	(0.58)	(-0.06)	(-2.80)	(-3.37)	(-0.11)	(-0.06)
Win Experience	-0.0458	-0.0258	0.0069	0.0093	-0.0019	-0.0023
	(-0.51)	(-0.26)	(0.42)	(0.56)	(-0.14)	(-0.17)
Gain × Loss Experience		0.0779**		0.0131*		-0.0002
		(2.39)		(1.77)		(-0.02)
Gain × Win Experience		-0.0372*		-0.0154**		0.0043
		(-1.66)		(-2.06)		(0.58)
Constant	19.2221***	19.2438***	2.3558***	2.3639***	1.4261***	1.4263***
	(49.53)	(49.33)	(38.43)	(38.04)	(27.10)	(28.02)
Observations	18,350,938	18,350,938	22,437,710	22,437,710	2,585,492	2,585,492
R-squared	0.146	0.146	0.100	0.100	0.128	0.128
Investor, stock, time FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes. The baseline specification is estimated separately within the holding-period groups shown in the header. Holding period is measured at the decision date. The dependent variable is an indicator for sale, multiplied by 100. Loss Experience and Win Experience count prior sales with realized returns below −5% and above +5%, respectively, relative to the investor's average purchase price. Gain indicates that the position is at a gain relative to its cost basis. Columns (1), (3), and (5) include both counts; Columns (2), (4), and (6) add both Gain interactions, as in Table 3, Column (5). The unit of observation is the investor–stock–day; the sample includes holdings on dates when the investor sells at least one security, from July 2013 to February 2016. All specifications include investor, stock, and time fixed effects. Standard errors are clustered by investor, time, and stock; *t*-statistics appear in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. Significance markers test coefficients against zero within each bin; they do not test differences between bins.

We re-estimate the full specification separately for investors at or below and above the median age of 36. Table 13 shows that the baseline disposition effect is about the same in both groups, 2.30 and 2.59 percentage points. Moreover, all four interactions are significant within their age groups and have the same signs as in the full sample. However, both experience interactions are more than twice as large in absolute value among younger investors. These results are consistent with the model and offer independent support for its account of experience: while age leaves the baseline effect roughly constant, it reduces the response to experience.

Table 13: Trading Experience by Investor Age

Dependent variable	Sale × 100			
Investor age	Age ≤ 36		Age > 36	
	(1)	(2)	(3)	(4)
Gain	2.3935***	2.2976***	2.5789***	2.5874***
	(4.20)	(3.28)	(5.20)	(4.91)
Loss Experience	0.0172***	0.0037	0.0180*	0.0124
	(3.92)	(0.72)	(1.85)	(1.11)
Gain × Loss Experience		0.0402***		0.0160***
		(3.62)		(2.66)
Win Experience	0.0329***	0.0425***	-0.0070	-0.0026
	(6.69)	(7.44)	(-0.78)	(-0.25)
Gain × Win Experience		-0.0354***		-0.0154**
		(-5.91)		(-2.55)
Constant	8.3169***	8.4253***	8.1570***	8.1998***
	(40.31)	(34.95)	(37.11)	(36.68)
Observations	19,396,148	19,396,148	24,253,718	24,253,718
R-squared	0.145	0.145	0.133	0.133
Investor, stock, time FE	Yes	Yes	Yes	Yes

Notes. The baseline specification is estimated separately for investors at or below and above the median age of 36. The dependent variable is an indicator for sale, multiplied by 100. Loss Experience and Win Experience count prior sales with realized returns below −5% and above +5%, respectively, relative to the investor's average purchase price. Gain indicates that the position is at a gain relative to its cost basis. Columns (1) and (3) include both counts; Columns (2) and (4) add both Gain interactions, as in Table 3, Column (5). The unit of observation is the investor–stock–day; the sample includes holdings on dates when the investor sells at least one security, from July 2013 to February 2016. All specifications include investor, stock, and time fixed effects. Standard errors are clustered by investor, time, and stock; t-statistics appear in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. Significance markers test coefficients against zero within each age group; they do not test differences between groups.

6 Conclusion

Using transaction records for 189,530 Chinese retail investors over the 2013–2016 boom and bust, we find that the baseline winner–loser selling gap narrows with prior realized gains and widens with prior realized losses. A two-standard-deviation increase in accumulated large losses is associated with a 0.57-percentage-point wider gap, 23% of the baseline effect of 2.48 points; the corresponding increase in large gains is associated with a 0.51-percentage-point narrower gap, 21%. We show that these effects cannot be explained by the tendency of experience to increase the number of positions in a portfolio, nor are they likely to be explained by overconfidence or extrapolation. We show that winning experience continues to reduce the disposition effect once the portfolio's own gain or loss is held fixed, and that the effect is present for both short holding periods of zero to two months, and longer holding periods between two and eight months. Finally, we show that experience matters more for younger than for older investors.

We account for these results with a model in which cued recall determines selling decisions. Investors are more likely to sell positions that come to mind, and the positions that come to mind are those closest to the cue, the stocks they own; the most stock-like positions, namely the ones with the most extreme returns, tend to be the ones sold. Because stock returns are highly skewed, these tend to be winners. Small returns are experienced on many things besides stocks, so a small winner or loser rarely comes to mind. As winning positions are sold, the stocks an investor owns become less associated with large gains, gains come to mind less often, and the disposition effect falls. As losing positions are sold, losses come to mind less often and gains more, and the disposition effect rises. While previous work on experiences focuses on beliefs, our channel relates directly to what comes to mind. The ability to recall the past, and what is recalled, shape decisions not only through beliefs, but more directly, depending on the nature of the decision with which the investor is faced. We look forward to future work disentangling these channels.

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Appendix A Additional Information

A1 Calculating Trading Returns

Transaction-Level Returns The percentage-return definition in Section 2.3 compares the sale price with the volume-weighted purchase-price reference for the shares sold. Let $P^s_{i,j,t}$ denote the sale price and $\bar{P}^b_{i,j,t}$ this purchase-price reference. The corresponding return is

$$r^s_{i,j,t} = \frac{P^s_{i,j,t} - \bar{P}^b_{i,j,t}}{\bar{P}^b_{i,j,t}}. \tag{19}$$

Multiplying the numerator and denominator by the number of shares sold expresses the same return as realized profit divided by the purchase cost assigned to those shares. Dividing instead by cumulative expenditure on all shares purchased gives a different measure when only part of the holding is sold. Additional purchases enter the reference price at their volume-weighted purchase cost. A complete liquidation followed by a new purchase begins a new holding.

Paper returns use the same purchase-price reference and replace the sale price with the opening price on the decision date. Paper returns are more often negative than positive but strongly right-skewed, with skewness of approximately 23. Section 4 considers how the distribution of return experiences affects recall.

Stock-Level Returns For a holding that is fully liquidated, total sale proceeds relative to total purchase expenditure are

$$R_{ij} = \frac{\sum_{\tau=t_0}^{t_1} q^s_{ij\tau} P^s_{ij\tau}}{\sum_{\tau=t_0}^{t_1} q^b_{ij\tau} P^b_{ij\tau}} - 1, \tag{20}$$

where t_0 is the first purchase date, t_1 is the date of complete liquidation, and $q^s_{ij\tau}$ and $q^b_{ij\tau}$ are the numbers of shares sold and bought on date τ. A quantity is zero on dates without the corresponding trade; prices are volume-weighted when there is more than one trade of the same type on a date. This is the net trading gain relative to total purchase expenditure, rather than a return adjusted for the timing of cash flows.

A2 Empirical Strategies for Quantifying Investor Experience

We construct trading episodes following Seru, Shumway and Stoffman (2010). An episode begins with a purchase and ends at the first partial or complete sale, whose realized return classifies the experience as a gain or loss. After a partial sale, the remaining shares begin

a new episode. These episodes define the experience counts used to relate past realized outcomes to current selling decisions.

A3 Mechanical Bias from Portfolio Composition

Experience counts and current selling decisions both depend on the evolution of an investor's portfolio. Portfolio size and the fraction of positions at a gain can therefore generate associations between experience and the winner–loser selling gap even when selling weights do not depend directly on experience. Under the rule with fixed relative selling weights, a portfolio with W winners and L losers has per-position sale probabilities $1.5/(1.5W + L)$ for winners and $1/(1.5W + L)$ for losers. The gap is $0.5/(1.5W + L)$. It decreases as the portfolio expands at a fixed winner share and increases as the winner share falls at a fixed portfolio size. The simulations assess the combined associations generated by portfolio growth, changes in composition, and the selling rule.

Portfolios evolve between sales. The effect of a sale on subsequent portfolio composition depends on intervening prices and purchases. Price changes can turn winners into losers or losers into winners, and new purchases change the set of positions available for sale. The composition immediately after a sale therefore need not persist until the next selling decision. Our simulation allows both prices and holdings to change between sales.

Within-investor calibration. Our empirical specifications include investor fixed effects. We therefore target within-investor correlations of experience with portfolio size and the held winner share. These moments discipline selected portfolio relationships; they do not reproduce the full empirical joint distribution.

Monte Carlo Assessment of Mechanical Bias

Selling weights do not respond directly to past experience. The simulation measures the experience interactions generated by portfolio dynamics and the regression specification under these rules, together with sampling variation.

Simulation design. We simulate 5,000 investors and repeat the exercise 200 times. The number of sell decisions per investor follows a log-normal distribution, rounded and restricted to between 1 and 600. This allows substantial heterogeneity in the number of sales and accumulated experience. Each investor starts with about five stocks and, after each sale, purchases new positions toward a target that rises over the trading sequence. We choose

this growth process to approximate the mean portfolio size and the positive within-investor correlations between experience and portfolio size in the data. A bounded decline in the market level over each investor's trading sequence helps approximate the negative correlations between experience and the held winner share. The targets are $+0.61$ and $+0.58$ for the size correlations and -0.26 and -0.16 for the winner-share correlations, for winning and losing experience, respectively. Across the three scenarios, the corresponding simulated correlations average 0.596–0.609, 0.599–0.608, -0.279–-0.241, and -0.138–-0.116, respectively, over the first 20 replications.

Between sales, prices evolve over three to seven simulated trading days through a log-normal process with moderate and rare large jumps. Jump parameters are calibrated to approximate the mean counts of winning and losing experiences at the 5% and 30% thresholds in Table 1. Algorithm 1 summarizes the data-generating process.

Algorithm 1. Data-generating process for one investor

1. Draw the number of sales $n = \text{round}\big(\text{LogNormal}(2.20,\ 1.48)\big)$, capped to $[1, 600]$.
2. Draw the starting size $k_0 = \max\big(3,\ \min(40, \text{round}(5 + \mathcal{N}(0, 0.6^2)))\big)$; open $K = k_0$ positions with purchase prices normalized to one and initial market prices $\exp(\mathcal{N}(0, 0.015^2))$. Draw the investor's market-decline amplitude a_i uniformly on $[0.048, 0.112]$.
3. For each sell event $\tau = 1, \ldots, n$:
   (a) If $K < 3$, replenish to k_0 positions.
   (b) Evolve prices for three to seven days: $p \leftarrow p \cdot \exp\big(\mathcal{N}(\mu, \sigma^2) + J^+_{\text{mod}} + J^+_{\text{big}} + J^-_{\text{mod}} + J^-_{\text{big}}\big)$, with independent moderate and rare-large jumps.
   (c) For $n > 1$, scale prices to the bounded market level $M(\tau) = \exp\big(-a_i(\tau-1)/(n-1)\big)$, applied incrementally relative to the previous level. Set $M(\tau) = 1$ when $n = 1$. This bounds the cumulative log-price shift by a_i.
   (d) Classify each holding as a gain or loss from its unrealized return.
   (e) Assign sell weights—**(A)** equal; **(B)** 1.5× for gains; **(C)** rule (B) $\times(1 + 0.8\ \min(|r|, 1)/0.10)$—and sell one position with probability proportional to its weight.
   (f) Record all current holdings: the sale indicator, the gain dummy, and the running counts WinExp and LossExp.
   (g) If the realized return is above 5% (below -5%), increment the winning (losing) experience counter. Apply the same rule at the 30% threshold.
   (h) Remove the sold position; set the target $= \min\big(k_0 + \min(0.22(\tau-1),\ 2.5),\ 40\big)$ and stochastically buy back toward it.

We consider three selling scenarios:

(A) **Random selling.** Each stock within the same portfolio is equally likely to be sold. This provides a benchmark with no preference for winners or losers.

(B) **Fixed relative selling weights.** Winners' selling weight is fixed at 1.5 times losers' weight, so within the same portfolio a winner has 1.5 times the sale probability of a loser. This ratio is constant across experience levels, while the probability difference depends on portfolio size and composition.

(C) **Fixed relative weights with V-shaped selling.** Motivated by Ben-David and Hirshleifer (2012), we additionally give positions with larger absolute returns greater selling weight. The weights do not respond directly to accumulated experience.

For each simulated dataset, we estimate an investor-fixed-effects analogue of Table 3, Column (5):

$$\begin{aligned}\text{Sale}_{i,j,t} \times 100 = \alpha_i + \beta_1 \,\text{Gain}_{i,j,t} + \beta_2 \,\text{LossExp}_{i,t} + \beta_3 \,\text{WinExp}_{i,t} \\ + \beta_4 \,\text{Gain}_{i,j,t} \times \text{LossExp}_{i,t} + \beta_5 \,\text{Gain}_{i,j,t} \times \text{WinExp}_{i,t} + \varepsilon_{i,j,t}\end{aligned}$$

We include investor fixed effects and cluster standard errors by investor. The empirical specification also includes stock and time fixed effects and uses three-way clustering. The simulation does not reproduce the common-stock and calendar-time structure of the empirical panel. Its estimates are therefore comparisons under calibrated portfolio dynamics, rather than estimates from an exact replication of the empirical design. The coefficients of interest are β_4 and β_5.

Results. Table 7 reports the mean simulated coefficient, the 95% simulation interval, and the tail fraction across 200 replications. Figure 4 plots the corresponding distributions at the 5% threshold. The tail fraction is the share of simulations producing a loss interaction at least as large as the empirical estimate or a win interaction at least as negative as the empirical estimate.

Interpretation. At the 5% threshold, the mean loss-side interaction under each calibrated scenario is negative or close to zero. The means are -0.0064 under the rule with fixed relative selling weights, -0.0068 under V-shaped selling, and $+0.0006$ under random selling, compared with the empirical estimate of $+0.0212$. At most five of the 200 replications in each scenario produce a coefficient as large as the empirical estimate.

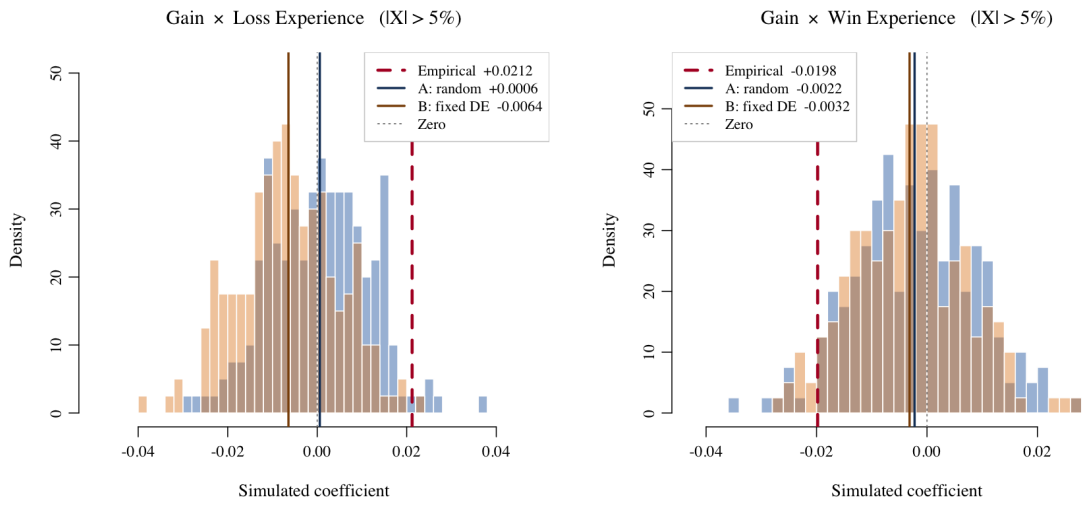


Figure 4: Distribution of Simulated Gain × Experience Coefficients ($|X| > 5\%$)

Notes. The two panels show the distribution of the Gain × Loss Experience (left) and Gain × Win Experience (right) coefficients across 200 Monte Carlo replications at the $|X| > 5\%$ threshold, under the calibration targeting the within-investor relationships between experience, portfolio size, and the held winner share described above. Blue histograms correspond to Scenario A (random selling); orange histograms to Scenario B (fixed relative selling weights); Scenario C (fixed relative weights + V-shape) is reported in Table 7 but not plotted here. Dashed red lines mark the paper's point estimates from Table 3.

The mean winning-experience interactions are −0.0022 under random selling, −0.0032 under the rule with fixed relative selling weights, and +0.0019 under V-shaped selling, compared with the empirical estimate of −0.0198. At most nine of the 200 replications in each scenario produce an interaction as negative as the empirical estimate. The negative mean interaction under the rule with fixed relative selling weights is about one-sixth of the empirical estimate in magnitude.

Equal sale probabilities within a portfolio do not require the pooled winner–loser selling gap to be zero. Portfolio size and composition vary together, so winners and losers can face different distributions of per-position sale probabilities in the pooled sample. The linear regression and investor fixed effects also affect the estimated interactions. The random-selling result does not separately identify the contribution of each feature.

At the 30% threshold, the mean simulated loss-side interactions remain negative, whereas the empirical estimate is positive at +0.0907. The empirical winning-experience estimate is −0.0634 and less precise, and the simulated distributions are wide.

The calibrated scenarios do not reproduce the empirical combination of a positive loss-side interaction and a substantially negative winning-experience interaction on average. The tail comparisons are descriptive because the simulated and empirical panels differ in size and dependence structure. They do not constitute a formal rejection of all mechanical

explanations or identify recall as the source of the empirical pattern.